balanced_leverage_max

equity strategy | family: equity-aggressive | status: research | derived from `balanced_leverage`

What this strategy does

balanced_leverage with bigger bets for more return: base unit 0.012, 6x leverage cap

Why it should work. Market-neutral earnings drift with enough gross exposure to matter; the neutrality is the hedge.

Design notes (from the strategy's source file).

Aggressive-sizing variants of the hand-built equity strategies.

Same signal, trim, tilt and neutralization as each base strategy; only the two dials that set how big and how levered the book is change (base_unit_fraction, gross-leverage cap = 6x). The multiplier for each was picked by sweeping 1.5x..12x the original base unit and taking the highest return subject to Sharpe >= 0.6 and max drawdown >= -40% (a worse drawdown is a fund-ending event, not a risk preference). The untouched 5%-of-ADV liquidity cap remains the real capacity constraint, and bites roughly four times as often at this sizing.

These are generated from the base strategy classes, so a change to a base strategy's weights flows through automatically.

parameter	value
base_unit_fraction	0.012
max_gross_leverage	6.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
beta_overlay	0.0

Performance

metric	balanced leverage max	market (CRSP VW)
Annualized return	12.15%	9.94%
Annualized volatility	12.20%	19.35%
Sharpe (NAV-pct)	1.00	0.59
Sortino	1.45	
Max drawdown	-36.5%	-55.5%
Calmar	0.33	
Final NAV ($10M start)	$93,270,642	
Worst day	-7.14%	
% positive days	51.5%	
Sharpe 1996-2001	1.53	

Sharpe 2002-2007	1.71	
Sharpe 2008-2013	-0.07	

Correlation of daily returns with the market: **-0.02**.

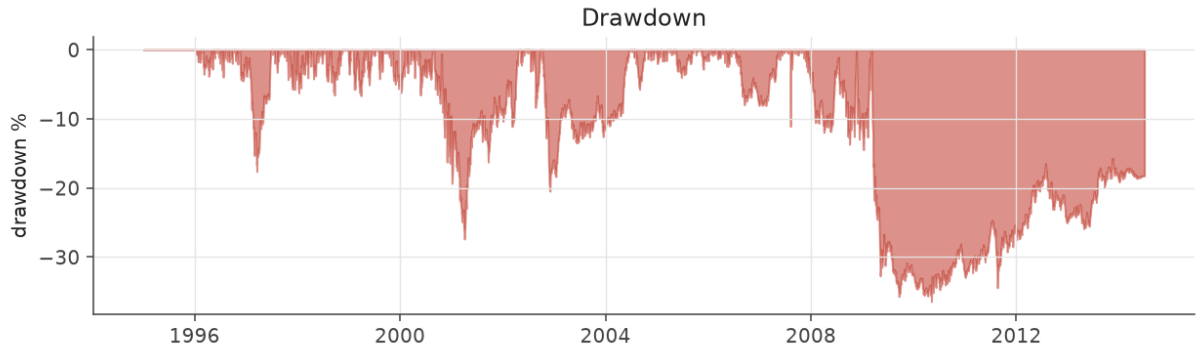
• best Sharpe among equity: `calm_rank_tilt` (1.81); this one: 1

• best return among equity: `market_boost_max` (0.15); this one: 0.121

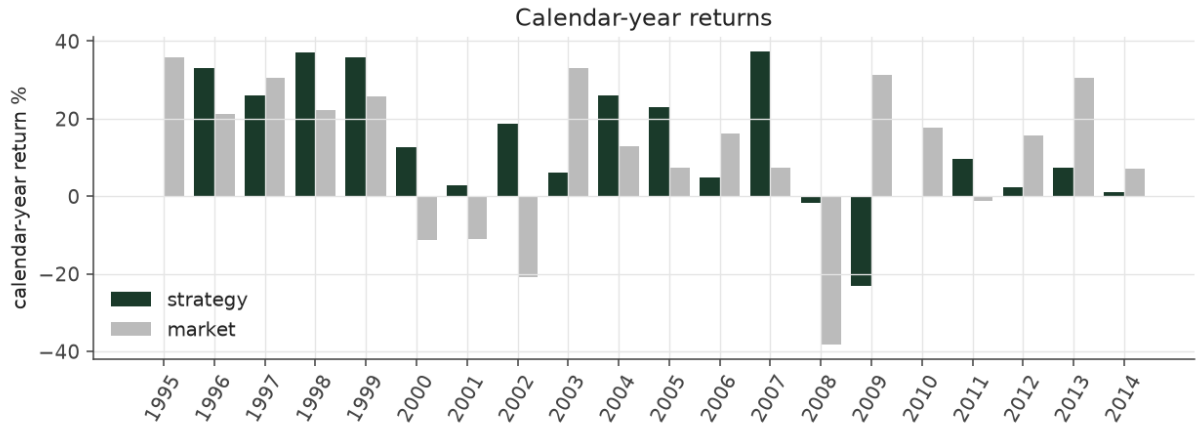
• best Calmar among equity: `calm_rank_tilt` (0.614); this one: 0.333



NAV (log scale)



Drawdown from running peak

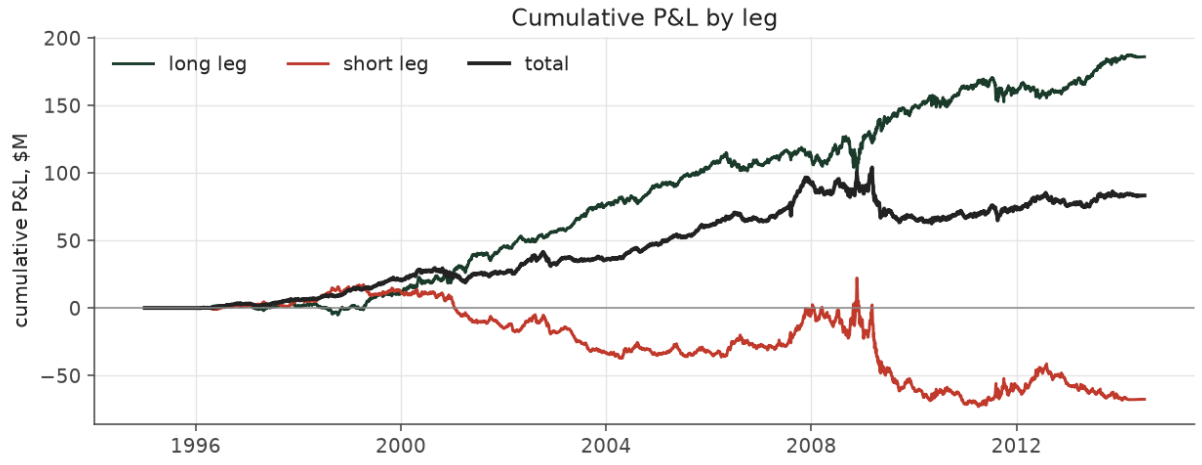


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	185.98	-67.62
P&L after costs ($M)	168.87	-85.60
of which common (average-event) component ($M)	174.40	-171.82

of which excess / PEAD component ($M)	11.58	104.20
excess minus costs ($M)	-5.54	86.22
gross return per $ traded (bp)	166.77	-59.88
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.06	-0.22

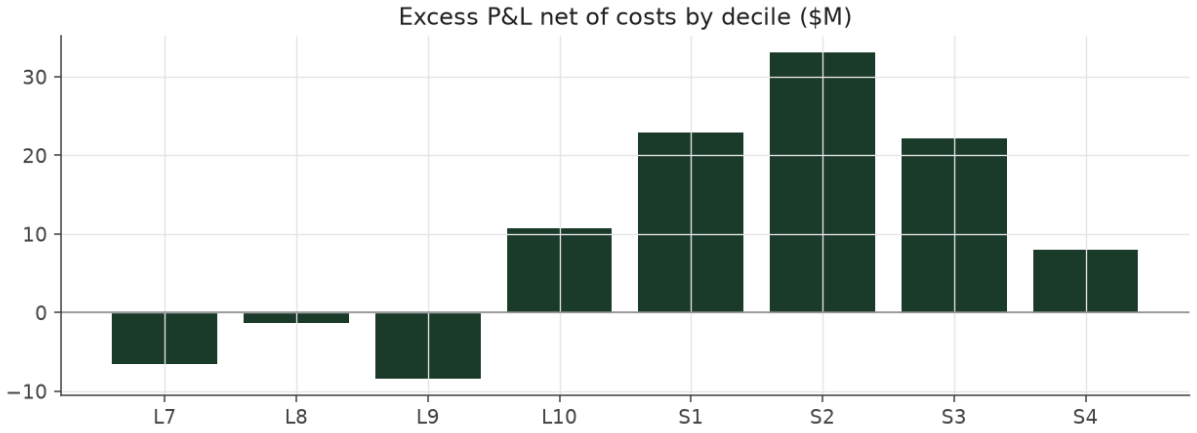
Daily P&L correlation between the two legs: **-0.77**.

• **The short leg loses money in absolute terms** (-67.6M before costs) while the long leg earns +186.0M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 174.4M / 171.8M. Stripping that common component, the surprise signal itself earned +11.6M on the long side and +104.2M on the short side.

• The two legs' daily P&L are -0.77 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.06 long / -0.22 short vs 1.00 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	7,718	1,052.78	11.28	-4.89	1.67	-6.56	-8.1%	-62.28
long	8	16,225	2,721.85	42.43	2.92	4.23	-1.31	-1.6%	-4.80
long	9	17,480	3,520.62	43.56	-3.12	5.32	-8.44	-10.5%	-23.97
long	10	16,939	3,857.22	71.60	16.66	5.90	10.76	13.3%	27.91
short	1	16,755	3,662.36	-27.10	28.88	5.92	22.95	28.5%	62.68
short	2	16,916	3,529.55	-24.03	38.62	5.54	33.09	41.0%	93.74
short	3	15,579	2,858.54	-23.05	26.72	4.57	22.15	27.4%	77.48
short	4	9,015	1,241.23	-11.41	9.98	1.95	8.04	10.0%	64.74



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	12,943	580.43	4.74	8.64	1.91	6.73	8.3%	115.97
2	22,307	2,425.20	15.93	23.56	5.94	17.62	21.8%	72.64

3	25,252	4,605.79	25.85	33.11	8.66	24.46	30.3%	53.10
4	27,425	6,685.72	17.20	24.57	9.76	14.81	18.4%	22.15
5	28,700	8,147.02	19.55	25.91	8.84	17.07	21.2%	20.95

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	11,934	272.18	5.63	7.59	1.36	6.23	7.7%	228.86
2	22,802	1,937.75	12.15	20.54	5.81	14.73	18.3%	76.01
3	25,740	4,782.62	26.45	37.71	9.56	28.14	34.9%	58.84
4	27,494	7,253.58	21.14	28.42	10.15	18.26	22.6%	25.18
5	28,652	8,196.33	17.81	21.40	8.20	13.21	16.4%	16.11

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,165	9,600.44	-14.15	34.77	14.06	20.71	25.7%	21.57
41-60d	25,072	4,938.28	27.47	18.98	7.84	11.14	13.8%	22.57
61-90d	34,592	6,224.84	23.02	26.13	10.15	15.98	19.8%	25.66
>90d	11,798	1,680.60	46.93	35.90	3.04	32.86	40.7%	195.50

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	32,056	3,008.84	26.65	30.54	4.72	25.82	32.0%	85.81
2002-07	39,668	7,889.54	61.23	69.57	12.19	57.38	71.1%	72.73
2008-13	44,903	11,545.77	-4.60	15.67	18.18	-2.51	-3.1%	-2.18

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	21,429	2,583.01	6.03	8.80	4.33	4.47	5.5%	17.32
Chems	2,571	1,235.93	7.33	9.13	1.67	7.47	9.3%	60.42
Durbl	2,800	1,290.38	2.69	4.60	1.78	2.82	3.5%	21.88
Enrgy	6,106	1,767.37	16.94	20.24	2.50	17.74	22.0%	100.38
Hlth	11,413	2,280.39	12.88	14.74	3.68	11.06	13.7%	48.52
Manuf	11,825	2,139.00	5.43	8.82	3.40	5.42	6.7%	25.33
Money	17,141	2,149.44	11.87	16.47	3.75	12.72	15.8%	59.20
NoDur	5,000	1,595.65	5.49	6.52	2.31	4.21	5.2%	26.40
Other	18,022	2,447.24	8.16	12.96	4.30	8.66	10.7%	35.37
Shops	11,767	2,235.81	13.10	15.47	3.56	11.91	14.8%	53.27

Telcm	3,932	1,386.74	-7.51	-4.96	1.87	-6.83	-8.5%	-49.22
Utils	4,621	1,333.20	0.86	2.98	1.96	1.02	1.3%	7.64

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	38,814	8,085.79	2.77	19.12	11.77	7.36	9.1%	9.10
low vol	39,075	6,848.95	38.30	46.85	11.52	35.33	43.8%	51.58
mid vol	37,173	7,280.81	41.91	49.60	11.41	38.19	47.3%	52.46

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	12.15%	-0.00%
0.5%	10.7	11.45%	0.70%
1.0%	21.5	10.65%	1.50%
2.0%	43.0	8.65%	3.50%
5.0%	107.4	-100.00%	112.15%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.